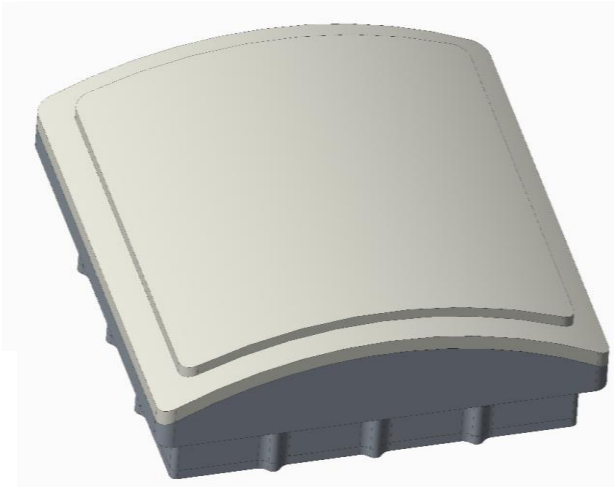
Thiagarajar Telekom Solutions (P) Limited

ANTIJAMMING ANTENNA SYSTEM – TTSLAJA-22



The TTSLAJA-22 is a 5 element Controlled radiation pattern array antenna operating at the L and S bands of IRNSS, GPS, and GLONASS constellations. The antenna provides good signal reception with wide beam hemispherical coverage pattern and good axial ratio. The mitigation of jammer signals is achieved by the implementation of beam forming and switching network creating deep nulls towards interference signals.

The array antenna with 4 auxiliary elements and one reference element has the capability of producing 3 simultaneous nulls to suppress three spatially distributed interference.

Key Features

- Rugged Architecture
- Simultaneous GPS L1 and GLONASS L1 protection
- Provides greater than 80 dB anti-jam protection

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The Compact design which meets the environmental conditions like missile applications the ruggedized enclosure with Aluminum Alloy (AA-2014) base and Teflon Radome. The TTSLAJA-22 is housed in a pre-tapped screw-hole mount, weather-proof enclosure for permanent installations.

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Technical Specifications

Electrical Specifications	
IRNSS L5	1176.45 MHz (±12MHz)
IRNSS S	2492 MHz (±8.25MHz)
GPS L1	1575 MHz (±10 MHz)
GPS L2	1227.26 MHz (±10 MHz)
GLONASS L1	1598 MHz-1606 MHz
GLONASS L2	1242 MHz – 1249 MHz
VSWR	≤ 2.0 :1
LNA Gain	28 dB ± 2 dB
Voltage Rating	28 V
Power Rating	20 W
Noise Figure	≤ 3.5 dB
Output Impedance	50 Ohms
Axial Ratio	< 3 dB
Lightening Protection	DC Grounding
Connector	SMA - (F)

Mechanical and Environmental Specifications- Antenna Unit	
Dimension (L X W X H)	150 x 150x 55 ±0.5mm (Excluding connector)
Dimension (L X W X H)	150 x 150x 78.5 ±0.5mm (Including connector)
Cavity material	Aluminium alloy- AA 2014
Radome material	PTFE Teflon
Weight	1350 ± 30 grams
Operating temperature	-40˚C to 80˚C

Anti-Jamming Performance: Protection Bands	
GPS L1	1575.42 ± 10 MHz
GLONASS L1	1602.42 ± 8 MHz
Interference Suppression	80 dB
No of Elements	5
No of Nulls	3

Environmental Stress Screening	
Mechanical Vibration	MIL-STD-810G / JSS 55555
Functional Shock	MIL-STD-810G / JSS 55555
Crash hazard	MIL-STD-810G / JSS 55555
Temperature and Altitude	MIL-STD-810G / JSS 55555
Humidity	MIL-STD-810G / JSS 55555
Temperature Shock	MIL-STD-810G / JSS 55555
Salt Fog	MIL-STD-810G / JSS 55555
Fungus	MIL-STD-810G / JSS 55555
Sand and Dust	MIL-STD-810G / JSS 55555
EMI & EMC	MIL-STD-461F

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